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VEXTRONICS

AGRICULTURAL & OUTDOOR ROBOTICS

WHAT IS AUTONOMOUS FARMING?



Concept:

- Using autonomous robots to perform farming actions

Opens up new possibilities:

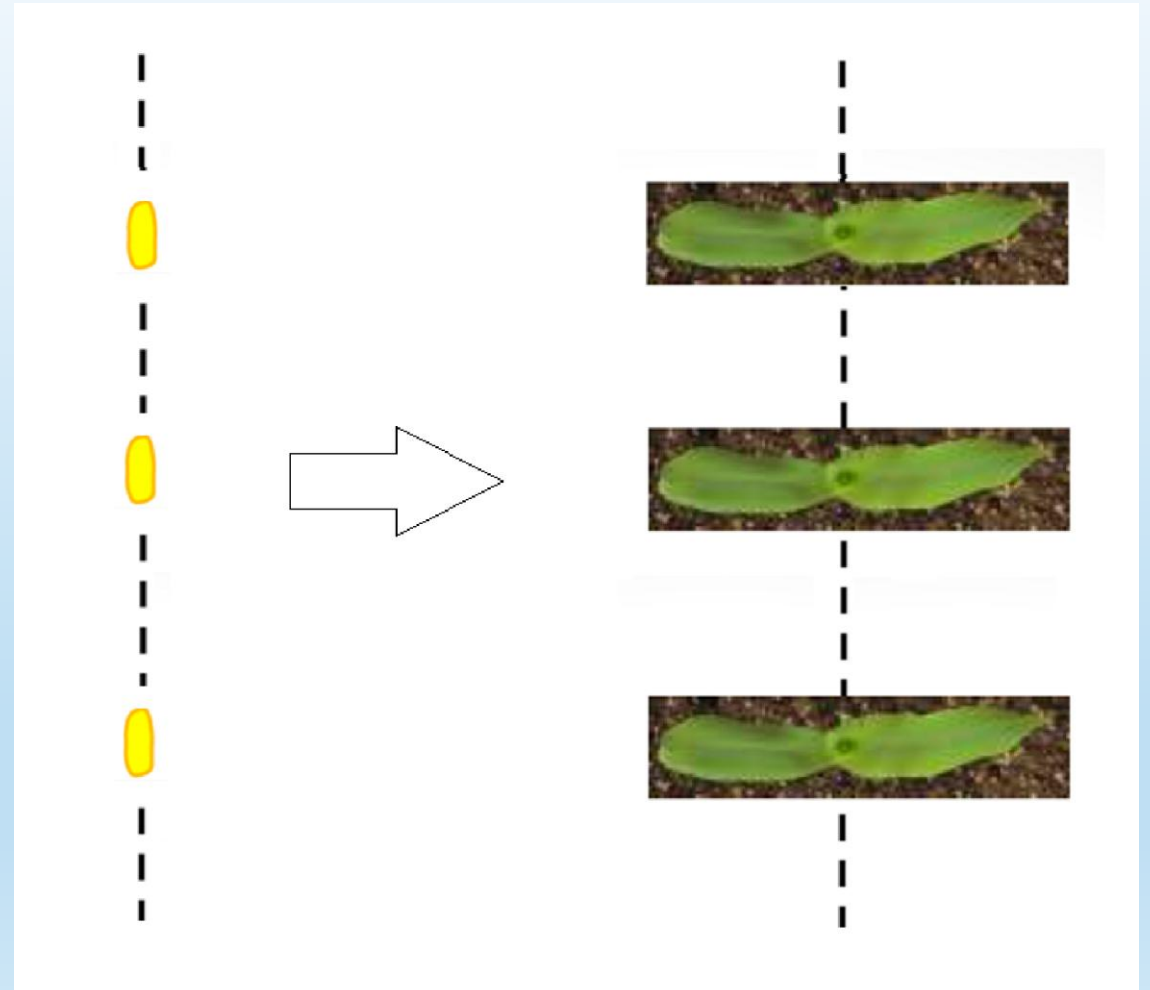
- More precision
- No more human errors
- Allows for higher speeds



Your challenge: Use robotics & AI to optimize planting of corn seeds.

THEORY: PLANTING CORN IN A PREDEFINED ALIGNMENT

Field Testing

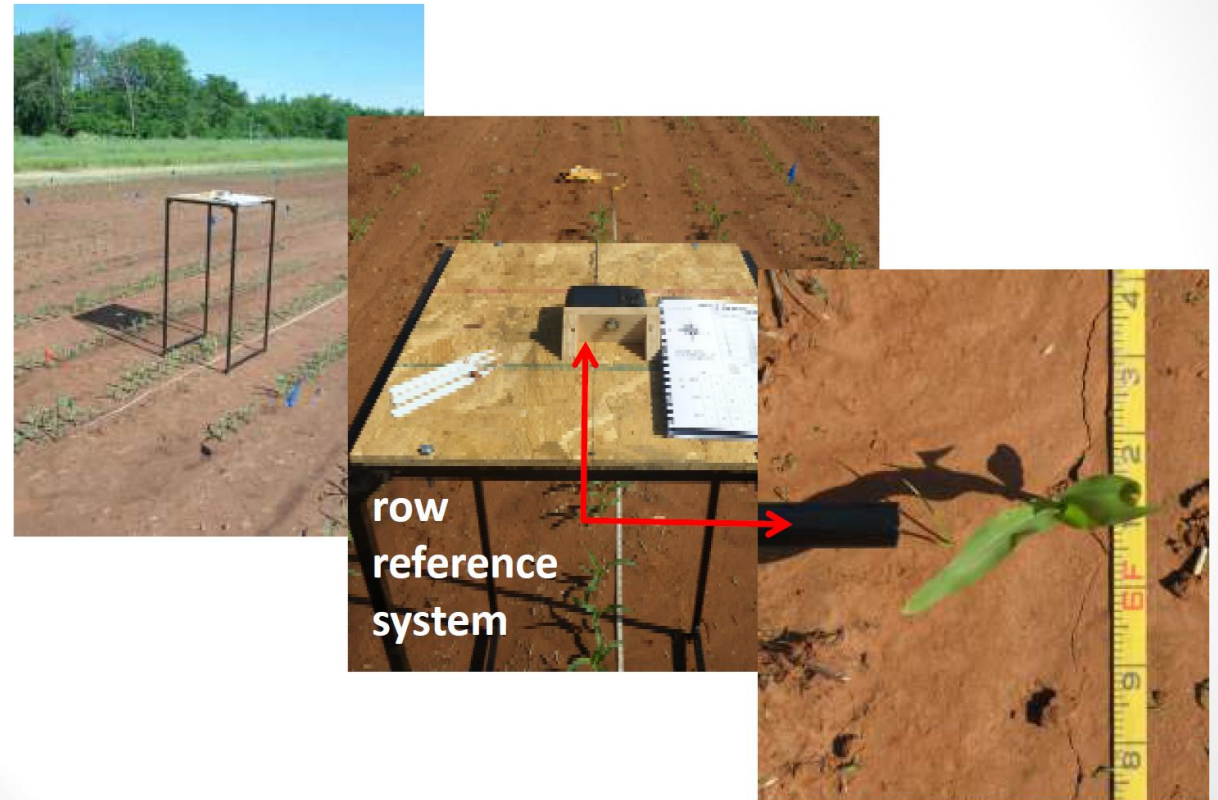


PROBLEM: HOW TO DETERMINE THE AVERAGE LEAVE ORIENTATION?

Your mission:

1. Create good quality plant data of corn seedlings.
2. Feed into AI object detection models.
3. **Determination of individual leave orientations.**
4. Show statistical results over the field.

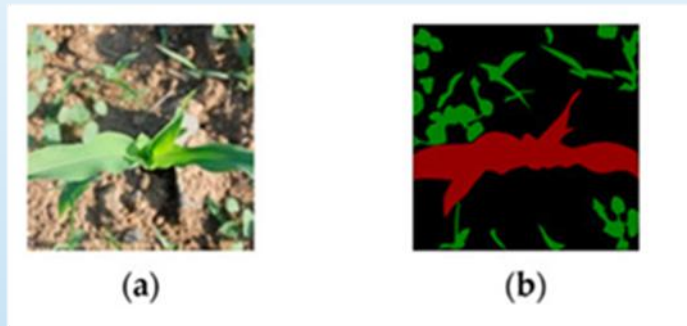
Collection of Field Data





DETERMINATION OF LEAF ORIENTATIONS USING AI TECHNIQUES

- Can we recognize the corn leaf shapes using Semantic Segmentation?



- How to calculate leaf orientation and plant center from the leaf shape?

